

Ben-Gurion University of the Negev
Faculty of Engineering Science
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Dept. of Electrical and Computer Engineering

Fourth-Year Engineering Project

Progress Executive Summary

MagViVit: Transformer-based spatiotemporal magnification for vital signs monitoring

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1. Project Background and Goals

The MagViVit project introduces a deep learning POC for continuous, non-contact monitoring of body temperature and respiration rate in preterm infants. To eliminate skin injuries (MARSI) and data gaps caused by traditional physical sensors, the system employs a touchless Dual-Branch architecture. This approach integrates CNNs for spatial feature extraction with Transformer Encoders for temporal modeling, aiming to achieve high-precision accuracy of MSE ≤ 0.2°C for temperature and MSE ≤ 2.0 breaths/minute for respiratory rate, to enhance clinical safety and neonatal care.

2. Accomplishments and Milestones

2.1. IRB Approval

Helsinki Committee (IRB) approval has been obtained to conduct a non-contact video monitoring study on preterm infants. This approval is a mandatory prerequisite for conducting research on human subjects in Israel, and it authorizes us to record preterm infants under real-world clinical conditions.

2.2. Clinical Data Acquisition System Deployment

Successful deployment of the data collection system was carried out in the NICU at Soroka Medical Center in Beersheba. As part of this phase, we visit the ward weekly to conduct comprehensive documentation and recording sessions. For data acquisition, we utilize a complete peripheral equipment suite, including RGB and NIR cameras from e-con Systems, interfaced with an NVIDIA Jetson AGX Orin edge processing unit. These synchronized sessions record preterm infants within their incubators. In parallel, custom mechanical arms and connectors were designed and 3D-printed, allowing the cameras to be securely stabilized above the incubators and medical monitors while ensuring zero interference with the medical staff's workflow.

2.3 Clinical Documentation Protocol

We established a structured clinical data documentation protocol, enabling the mapping of physical video files to medical metadata, including red blood cell count, skin tone, gestational age, weight, and other parameters. This protocol is essential for ensuring the quality and reliability of the dataset used for model training.

2.4 Hardware Acquisition

All required hardware components were purchased and integrated: cameras (RGB + NIR), an NVIDIA Jetson AGX Orin edge processing unit, a 1TB NVMe SSD, and USB3 interface adapters. Additionally, a GStreamer-based recording pipeline was developed for synchronized, frame-loss-free recording at a rate of 30fps.

2.5 Development of an algorithmic preprocessing pipeline

Monitor ROI extraction algorithm: a complete pipeline for extracting Ground Truth data from medical monitor screens, including camera calibration, lens undistortion, contour-based screen detection, KLT optical flow for feature tracking, RANSAC-based homography estimation, corner smoothing using a Kalman filter with Chi-Square outlier rejection, and finally, perspective rectification of the screen content.
Furthermore, the object detection annotation process for training data has been initiated: we have begun labeling two classes (face, face+torso) using CVAT and Docker, utilizing the YOLO format for the initial detection stage.

3. Future Tasks

Hardware and Documentation:

- Weekly recordings at the hospital are continuing to expand the dataset.

Algorithms:

- Respiratory Rate Branch: Integration of CNN and Video Motion Magnification (Autoencoder-based) algorithms.
- Temperature Branch: Training a UNet architecture for NIR-to-IR estimation mapping.
- Integration: Construction and training of the Dual-Branch Transformer architecture.
- Validation: Performance evaluation against Ground Truth (using MAE and RMSE metrics).